

# ACTUARIAL RESERVING MEMORANDUM

*Estimated Unpaid Loss Reserve -- Commercial Auto Liability*

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|----------------|--------------------------------------------------|-------------------|---------------------------|
| <b>TO:</b>     | Case Study File / Portfolio Reviewer             | <b>DATE:</b>      | September 2026            |
| <b>FROM:</b>   | Atishya Jindal (Reserving Analyst, illustrative) | <b>LOB:</b>       | Commercial Auto Liability |
| <b>ENTITY:</b> | State Farm Mut Grp (GRCODE 1767)                 | <b>EVAL DATE:</b> | 12/31/1997                |

**RE:** Chain-Ladder & Bornhuetter-Ferguson unpaid loss reserve estimate for Commercial Auto Liability

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## 1. Purpose

This memorandum documents an independent estimate of the unpaid loss reserve for the Commercial Auto Liability line of business, using publicly available NAIC Schedule P data. The purpose is to demonstrate the core reserving toolkit used across P&C actuarial functions -- the Chain-Ladder and Bornhuetter-Ferguson deterministic methods, plus a stochastic bootstrap to quantify the uncertainty around the point estimate -- and to compare the resulting estimates against the company's own booked reserve.

## 2. Data & Scope

- Source: CAS (Casualty Actuarial Society) Loss Reserving Database, compiled from NAIC Schedule P Part 3-4 statutory filings.
- Entity: State Farm Mutual Automobile Insurance Group (NAIC Group Code 1767).
- Line of business: Commercial Auto Liability, net of reinsurance.
- Experience period: 10 accident years (1988-1997), each evaluated at 12-month intervals through 10 years of development (i.e. through calendar year-end 1997).
- All figures are in \$000s (thousands of US dollars) unless noted.

## 3. Methodology

### 3.1 Chain-Ladder (Development Factor) Method

Volume-weighted average age-to-age factors were calculated from the cumulative paid-loss triangle and compounded into cumulative development factors (CDFs) to ultimate. A tail factor of 1.010 was assumed beyond 120 months of development, reflecting the judgment that Commercial Auto claims are largely, but not entirely, settled by that point. Each accident year's paid-to-date loss was multiplied by its corresponding CDF to project an ultimate loss; the reserve is the difference between projected ultimate and paid-to-date.

### 3.2 Bornhuetter-Ferguson (BF) Method

The Chain-Ladder method is unstable for immature accident years -- a single early, atypical payment can swing the projected ultimate wildly, because the CDF applied is large (over 3x for 1997, which has only one year of development). The BF method addresses this by using an a priori expected loss ratio (65.0%, selected from mature-year Chain-Ladder-implied ratios of ~68.5%, discounted slightly for conservatism given rising loss-cost trend) to independently estimate expected ultimate losses, then applies only the % UNREPORTED portion ( $1 - 1/\text{CDF}$ ) of that expectation as the reserve add-on. This blends the Chain-Ladder's own signal with a stable prior, weighted automatically toward the prior for the least mature years.

### 3.3 ODP Bootstrap (Reserve Range)

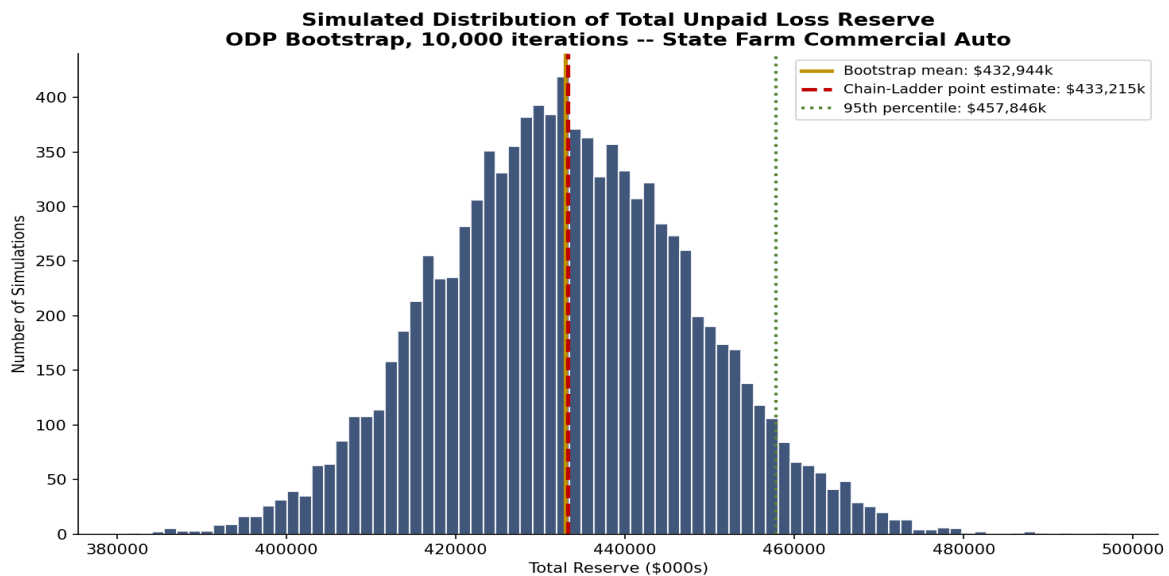
Both methods above produce a single point estimate, which understates the real uncertainty in the reserve. An Over-Dispersed Poisson (ODP) bootstrap (England & Verrall, 2002) was run with 10,000 iterations to convert the Chain-Ladder point estimate into a full probability distribution. Parameter uncertainty was captured by resampling the model's own Pearson residuals to generate thousands of alternate claim histories and refitting development factors to each; process uncertainty was layered on top by simulating each future payment as a Gamma-distributed random draw calibrated to the data's own variance. This is conceptually the same Monte Carlo logic used to simulate a P&L distribution for VaR in a trading book -- here the 'position' being resimulated is the insurer's unpaid claim liability.

## 4. Results

### 4.1 Reserve Estimate by Accident Year (\$000s)

| Accident Year | Paid-to- Date | Chain-Ladder Reserve | BF Reserve | CL Ultimate Loss Ratio |
|---------------|---------------|----------------------|------------|------------------------|
| 1988          | 193,499       | 1,935                | 1,843      | 68.2%                  |
| 1989          | 199,997       | 5,158                | 5,049      | 66.4%                  |
| 1990          | 224,078       | 7,407                | 6,790      | 70.9%                  |
| 1991          | 204,911       | 8,604                | 8,712      | 64.2%                  |
| 1992          | 208,008       | 12,205               | 12,316     | 64.4%                  |
| 1993          | 219,345       | 21,587               | 20,724     | 67.7%                  |
| 1994          | 212,446       | 36,299               | 36,023     | 65.5%                  |
| 1995          | 190,974       | 63,828               | 64,928     | 63.9%                  |
| 1996          | 143,590       | 102,468              | 110,063    | 60.5%                  |
| 1997          | 75,827        | 173,723              | 183,946    | 61.4%                  |
| Total         | 1,872,675     | 433,215              | 450,395    | 63.9%                  |

### 4.2 Bootstrap Reserve Distribution (10,000 simulations)



| Statistic              | Value                   | Interpretation                                                 |
|------------------------|-------------------------|----------------------------------------------------------------|
| Bootstrap mean reserve | \$432,944k              | essentially matches the \$433,215k Chain-Ladder point estimate |
| Standard deviation     | \$14,972k               | coefficient of variation = 3.5%                                |
| 5th / 95th percentile  | \$408,651k / \$457,846k | 90% confidence interval                                        |
| 99th percentile        | \$468,112k              | extreme-tail capital scenario                                  |

## 4.3 Comparison to Company's Booked Reserve

| Basis                                               | Amount                             |
|-----------------------------------------------------|------------------------------------|
| Company's booked (posted) statutory reserve, YE1997 | \$360,657k                         |
| Chain-Ladder reserve estimate                       | \$433,215k (+\$72,558k vs. booked) |
| BF reserve estimate                                 | \$450,395k (+\$89,738k vs. booked) |
| Bootstrap 90% CI                                    | \$408,651k -- \$457,846k           |

Both independent methods -- and the entire 90% bootstrap confidence interval -- sit meaningfully above the company's own booked reserve. Even the low end of the simulated range (\$408,651k) exceeds the booked amount by roughly \$48,000k. This is a real, recognizable reserving finding (a 'reserve deficiency' signal), not an error in either calculation: booked reserves reflect a company's own internal judgment, case-level claims adjuster estimates, and often a degree of conservatism or smoothing that a purely mechanical, external recalculation from aggregate triangle data cannot see. In practice this gap would trigger, not conclude, further investigation.

## 5. Judgment Calls & Sensitivities

- Tail factor (1.010): if the true tail is instead 1.030, the CL reserve rises by roughly 2% of the AY1988-1990 ultimates -- a modest but non-trivial effect given tail factors are the hardest development-factor assumption to support with only 10 years of data.
- A priori loss ratio (65.0% for BF): this is the single most influential BF assumption for the least mature years. Raising it to the mature-year average of 68.5% would raise the 1997 BF reserve by roughly \$13,000k.
- Volume-weighted vs. simple average age-to-age factors: volume-weighting (used here) is less sensitive to a single small, erratic accident year than a simple average would be -- appropriate given this triangle's largest accident years are also its most recent.
- The bootstrap's 3.5% coefficient of variation is on the low end of typical Commercial Auto benchmarks (often 5-15%), consistent with this being a very large, high-volume, unusually smooth triangle (State Farm's book) -- a smaller or newer insurer's book would show materially more reserve volatility.

## 6. Limitations & Reliances

- Paid-loss triangles only were used for the primary estimate; an incurred-loss (case + paid) Chain-Ladder was not cross-checked in this exercise, and in practice would be a required second opinion.
- No adjustment was made for known changes in claims settlement practices, mix of business, or large/catastrophic claims that could distort historical development patterns.
- This is an illustrative, external analysis built from public aggregate data for portfolio/demonstration purposes; it does not reflect internal claims files, case reserves, or company-specific book knowledge a real reserving actuary would have access to.

## 7. Files in This Analysis

| File                      | Description                                                                 |
|---------------------------|-----------------------------------------------------------------------------|
| Loss_Reserving_Model.xlsx | Live Chain-Ladder and BF calculations, all formulas auditable and editable. |

| File                                             | Description                                                                             |
|--------------------------------------------------|-----------------------------------------------------------------------------------------|
| Reserve_Bootstrap.py                             | Python ODP bootstrap simulation (10,000 iterations) producing the reserve distribution. |
| reserve_distribution.png / bootstrap_results.csv | Chart and raw simulation output referenced in Section 4.2.                              |
| Actuarial_Memo.pdf                               | This document.                                                                          |

Prepared as a portfolio project applying CAS/IFoA reserving fundamentals (Chain-Ladder, Bornhuetter-Ferguson, and stochastic reserving via bootstrap) to real, publicly available NAIC Schedule P data. Methodology follows standard actuarial practice as described in the CAS Statement of Principles Regarding P&C Loss and LAE Reserves and England & Verrall (2002), 'Stochastic Claims Reserving in General Insurance.'